Pre-analysis Plan

Research Team ID:

Brief Summary of the Project

The analysis outlined in this pre-analysis plan (PAP) is part of an international collaborative project involving approximately 150 researchers and research teams (hereafter R&RT). Each R&RT will test a predefined set of four hypotheses concerning how having daughters influences behavior and preferences, using data from the German Socio-Economic Panel (SOEP). All R&RTs are required to develop a PAP and provide analysis code for Stata. They are randomly assigned to one of two groups: the *data group*, which receives access to a 5% sample of the data for the period 2000-2020 while preparing their PAP, and the *no-data group*, which receives the sample only after submitting their PAP.

The overarching goal of the collaborative project, as described by the project coordinators, is "to gain insights into both the scientific process as well as the impact of having daughters on various outcomes. They [the project coordinators] will study variation in analytical choices and findings, and write a metascience paper on the topic."[1] No further details are provided to the R&RTs.

Group Assignment and Data Access

RT was randomly assigned to the **data group** on May 23, 2025, after completing the first of three required surveys and submitting their data use agreement for the SOEP. RT received access to the 5% sample of the data on May 25, 2025.

Hypothesis

The four pre-defined Hypothesis are:

H1: Having daughters leads to stronger support for gender equality (as measured by attitudes).
H2: Having daughters leads to more left-leaning political preferences.
H3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
H4: Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

R&RTs are asked to consider these hypotheses as causal hypotheses when developing their analysis.

[1] See https://www.manydaughters.com/

Sample

The SOEP provides a representative longitudinal sample of private households in Germany. This sample will be restricted to **adult respondents**, i.e. individuals aged 18 and older, residing in **private households** with complete information on the type of residence (that pertains to codes 1 and 2 of variable `pop`). We will thereby exclude respondents who reside in institutional residences (e.g., student dormitory, old age home, hotel). The sample will further be restricted to respondents with **successful interviews** (code 10 in the variable `netto`). This will exclude individuals who did not successfully complete the individual survey and for whom essential information required for our analysis is therefore missing (e.g., political preferences). We will combine information from several SOEP data sets in the analyses:

- `ppathl` (includes information about survey status, socio-demographic information and type of residence)
- `pl` (includes information about attitudes and behaviors)
- `biobirth` (includes information about respondents' fertility histories)

To comply with the requirement to test all four hypotheses as causal hypotheses, we will further restrict the sample to **respondents with at least two observations** on the variables relevant to the analysis of each respective hypothesis. For **Hypothesis 3**, we will further restrict the sample to respondents who **live with a partner who also participates in the SOEP**. Since this hypothesis focuses on gender equality, only **male-female couples** will be included in this analysis.

Variables and operationalization

Dependent variables

Gender equality as measured by attitudes

Unfortunately, the SOEP does not include repeated measures of attitudes on gender roles or gender equality. Research shows that people who hold post-materialist values are less likely to endorse traditional gender role attitudes (e.g., Hayes et al. 2000; Ben-Nun Bloom et al. 2017). We will therefore approximate pro-gender-equality attitudes via Inglehart's postmaterialism index. Respondents are asked to rank four political goals: "You can't have everything at once - and that applies to politics, too. In the following, we will state four possible goals that politicians might pursue. If you had to choose, which of these goals would be most important? Please rank them in order of importance starting with the first?" The four goals are: "Peace and order" (`plh0054`), "Citizen influence" (`plh0056`), "Fight against inflation" (`plh0058`), and "Freedom of speech" (`plh0061`). The goals "Citizen influence" and "Freedom of speech" are classified as postmaterialist, while "Peace and order" and "Fight Against Inflation" represent materialist goals. Respondents will receive a score of 1 if their top-ranked goal is postmaterialist, and 0 otherwise. If their second-ranked goal is postmaterialist, they receive an additional 0.5; otherwise, they receive 0. This will result in a postmaterialism index ranging from 0 (full materialists) to 1.5 (full

postmaterialists), with intermediate values of 0.5 and 1 representing mixed types.[2] The postmaterialism-item battery is available for the survey years 2006 and 2016. To model within-individual changes in postmaterialist values over time, we will construct a difference variable that captures the change in postmaterialist values between the survey wave 2016 and the preceding wave 2006. Higher values will indicate a change towards more postmaterialist value attitudes.

Left-leaning political preferences

Left-leaning political preferences will be captured using the SOEP item: "In politics, people often talk about 'left' and 'right' to characterize different political attitudes. Thinking about your own political views, where would you place yourself?" (`plh0004`). Respondents rated their position on a scale from 0 ("far left") to 10 ("far right"). To align with Hypothesis 2, which states that "having daughters leads to more left-leaning political preferences," we will reverse-code the scale so that higher values indicate more left-leaning attitudes. This item is available for the survey years 2005, 2009, 2014 and 2019. To model within-individual changes in political preferences over time, we will construct a difference variable that captures the change in political orientation between each survey wave t and the preceding wave t−1 for the years with available data. Higher values indicate a change towards a more left-leaning political stance. In order to remove influential cases, respondents whose political stance differs by more than 5 points from any one t-1 to t will be coded as missing.

Gender equality in the household as measured by behaviors or outcomes

We will assess gender equality within households as the absolute difference in total hours of labor (encompassing paid and unpaid work) performed by the male and female partners. We will use self-reported daily time spent on weekdays for both paid and unpaid tasks. The unpaid work variables include:

- running errands (`pli0040`)
- caring for other persons in need of care (`pli0046`)
- performing household chores (`pli0043_h`)
- engaging in repair work (`pli0049_h`)
- child care (`pli0044_h`)

Paid work will be captured through weekly working hours (`plb0186_h`), which will be divided by five to approximate daily working hours. All time use variables will be coded in hours, and we will recode answers larger than 24 as missing values. From these variables, we plan to construct a composite time use measure by summing all daily hours spent on paid and unpaid tasks. To capture the gender gap in time allocation, we will subtract the total time use reported by the female partner from that of the male partner. Finally, we will take the absolute value of this difference to capture the inequality in total labor contributions in the household. To capture gender *equality* in the household, we will invert this variable so that negative values represent stronger inequality and the maximum value of zero indicates full equality in the total labor contributions within the

[2] During the analysis, we found some respondents had erroneously chosen both postmaterialist goals as most important. These were recoded to 1.5.

household. To model within-individual changes in dyadic time allocation over time, we will construct a variable that captures the difference the gender gap in the total labor contributions between each survey wave t and the preceding wave t−1. Time use information is available for each survey year within our observation period (2000-2020). Higher scores will represent a smaller gender gap in total labor contributions.

Independent variables

Having daughters

To indicate whether a respondent has at least one daughter, we will construct a dummy variable using information on the birth year and sex of up to 19 biological children. First, we will identify all children born on or before the survey year using the birth year variables (`kidgeb01–kidgeb19`), flagging the case if the child was born in or before the survey year and 0 otherwise. Second, we will use the child sex variables (`kidsex01–kidsex19`) to identify daughters, coding female children as 1 and all others (male or missing) as 0.

We then generate a count variable that captures the number of daughters born by the survey year by multiplying the indicators for valid births with the corresponding daughter indicators. Finally, we will create a binary variable coded 1 if the respondent has at least one daughter (including households with daughters only or both daughters and sons), and 0 otherwise (including households with sons only, no children, or missing information).

Sex of the respondent

Sex (variable: `sex`) will be recorded as male (value=1) or female (value=0, reference category) based on information entered by interviewers in the address protocol prior to administering the questionnaire.

Control variables

Although covariates should ideally be associated with both the independent and dependent variable, and to the best of our knowledge the likelihood of having daughters in Germany is not systematically shaped biologically or socially by sociodemographic factors or gender-selective practices, we will nonetheless include two covariates in all models: *sex* and *age*. Research consistently shows that women tend to hold more egalitarian gender role attitudes and are politically more left-leaning (e.g. Kenny & Langsæther 2023; Langsæther et al. 2025). As Hypothesis 4 requires sex as a moderator, we will thus include it in all analyses to ensure consistency.

Research further shows that younger individuals tend to hold more egalitarian gender role attitudes and are generally more politically left-leaning than older individuals (e.g. Kenny & Langsæther 2023). As the likelihood of having a child, and thus having a daughter, increases with age and has during the observation period been steadily increasing by about 0.1 years annually (Destatis 2025a, 2025b) we will further control for age in all four analyses. The variable age will

be calculated by subtracting the birth year (`gebjahr`) from the survey year (`syear`). Respondents who are younger than 18 will be coded as missing.

Analysis strategy

H1: We will estimate a linear regression model (OLS) of the **postmaterialism change variable** on the existence of a daughter, the respondent's sex, and the respondent's age.
H2: We will estimate a linear regression model (OLS) of the **left-leaning change variable** on the existence of a daughter, the respondent's sex, the respondent's age, and the survey year.
H3: We will estimate a linear regression model (OLS) of the **equality in total labor contributions change variable** on the existence of a daughter in the household (i.e. if either the female or the male partner indicated to have a daughter), both partners' age, and the survey year.
H4: We will estimate a linear regression model (OLS) of the **postmaterialism change variable** on the existence of a daughter, the respondent's sex, the interaction between the existence of a daughter and the respondent's sex, and the respondent's age.

We will apply **robust standard errors** for H1 and H4 to account for heteroscedasticity in the residuals if necessary. For H2 and H3, we will use **cluster-robust standard errors** (cluster variable: `pid`) to account for the clustering of observations within respondents. All analyses will be **weighted** using standard individual weights (`phrf`).

Missing values will be deleted listwise within each hypothesis. This has two implications: First, due to missing data, the representativeness of the sample may be compromised. Second, different hypotheses may rely on different subsets of respondents, which might limit the comparability across the hypotheses.

All analyses are weighted using individual weights. While household weights would be more appropriate for Hypothesis 3 (where clustering within household level would also be more appropriate) these weights were not included in the datasets provided by the project team.

Interpretation of results

Two conditions must be met to determine whether the results support the respective hypothesis: First, the sign of the unstandardized regression coefficient $b_{daughter}$ must be positive to be in line with the assumptions made in Hypotheses 1 to 4. Second, statistical significance will be assessed using the *t*-statistic. If the absolute value of *t* exceeds 1.96 (corresponding to a 5% significance level in a two-tailed test[3]), the coefficient is considered significantly different from zero. If both the sign and statistical significance are in line with the hypothesis, the result is interpreted as supporting it.

[3] Under the assumption that the full SOEP sample exceeds 200 observations.

References

Ben-Nun Bloom, P., Gilad, S., & Freedman, M. (2017). Does exposure to other cultures affect the impact of economic globalization on gender equality?. International Political Science Review, 38(3), 378-395.

Destatis, Statistisches Bundesamt (2025a): Live births: Germany, years, age of the father, marital status of the father. Genesis Code: Code: 12612-0006. Under: https://www-genesis.destatis.de/datenbank/online/url/9e434ba5, last accessed: 12.07.2025.

Destatis, Statistisches Bundesamt (2025b): Live births: Germany, years, age of the mother, live birth order. Genesis Code: Code: 12612-0005. Under: https://www-genesis.destatis.de/datenbank/online/url/163ccbc6, last accessed: 12.07.2025.

Hayes, B. C., McAllister, I., & Studlar, D. T. (2000). Gender, postmaterialism, and feminism in comparative perspective. International Political Science Review, 21(4), 425-439.

Kenny, John, and Peter Egge Langsæther. 2023. “Environmentalism as an Independent Dimension of Political Preferences.” *European Journal of Political Research* 62(4):1031–53. doi:10.1111/1475-6765.12549.

Langsæther, Peter Egge, and Carl Henrik Knutsen. 2025. “Are Women More Progressive than Men? Attitudinal Gender Gaps in West European Democracies.” *International Political Science Review* 46(3):442–53. doi:10.1177/01925121241280069.